

— Montreal, June 15–19 —

DLMI2026

Summer School on Deep Learning for Medical Imaging 7th edition

Fonds de recherche
Nature et
technologies

Québec 



Centre hospitalier
universitaire
de Sherbrooke



Welcome message

The Organizers and Sponsors welcome you to the seventh edition of the Summer School on Deep Learning for Medical Imaging (DLMI)

We hope you will enjoy the event and your stay in Montreal

Sponsors:



Why the Summer School ?

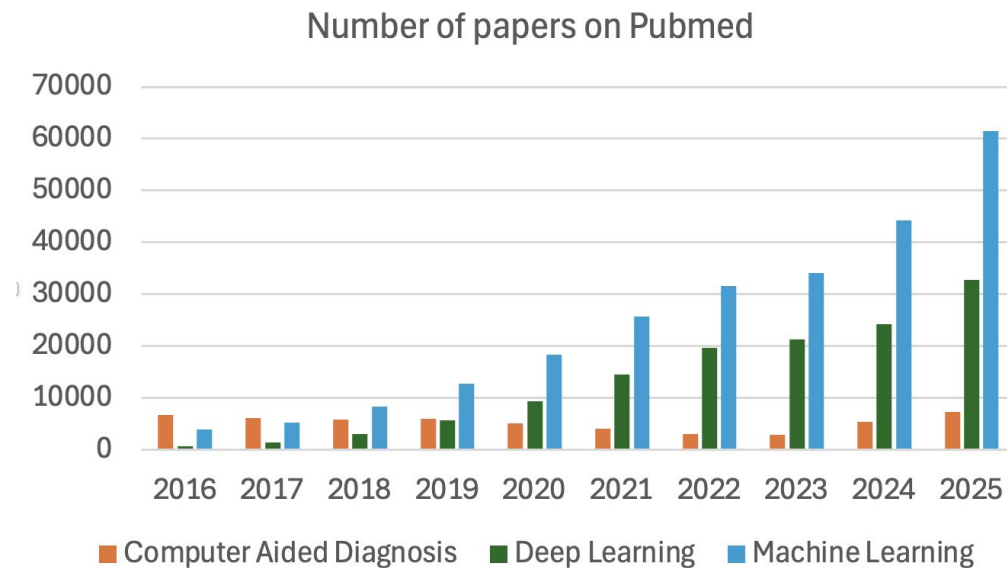
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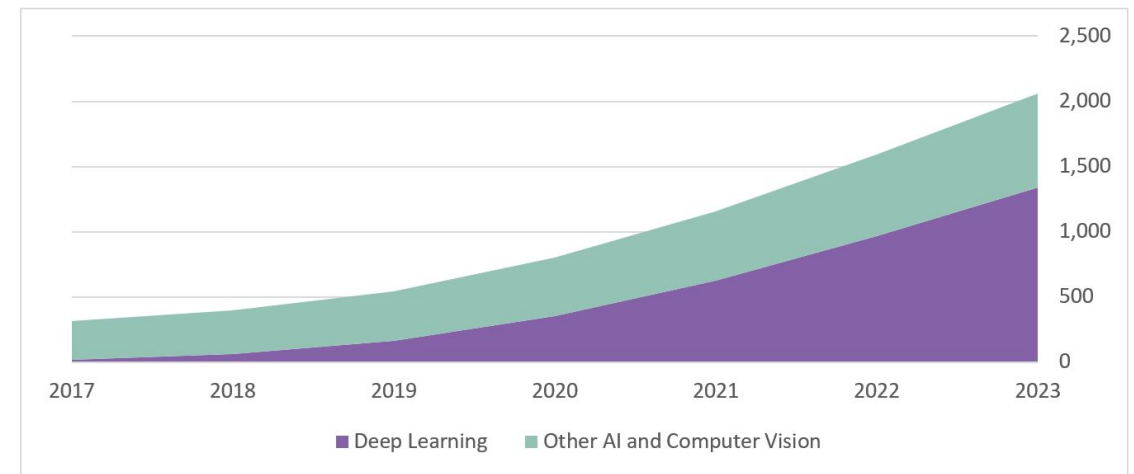
Introduction

Why the Summer School ?

- AI and deep learning have become an essential part of medical imaging



World Market for AI-Based Medical Image Analysis Software by Algorithm Type
Revenue Forecast (\$m)



Source: Signify Research

Jul-18



Introduction

What will I learn ?

Basics of deep learning:

- **Parts 1-2:** Perceptron and MLP, stochastic gradient descent, learning rate, logistic regression, activation function, regularization (L1/L2/dropout/early stopping), etc.
- **Part 3:** Weights initialization, backward propagation, batch size, convolution neural nets (CNN), pooling, etc.



Pierre-Marc Jodoin



Christian Desrosiers



Introduction

What will I learn ?

Advanced concepts in deep learning 1:

- Recurrent neural networks (RNNs), Attention, Transformers, ViT, etc.



Nicolas Thome

Advanced concepts in deep learning 2:

- Foundation models, language-image pre-training, parameter efficient fine-tuning, prompt learning.



Jose Dolz



Introduction

What will I learn ?

Key topics in DL and medical imaging:

1. CNN architectures and applications
2. Generative and adversarial methods
3. Uncertainty and explainability
4. Typical medical imaging issues
5. Semi and weakly supervised deep learning
6. Privacy protection in medical imaging



Michaël Sdika



Valentine
Wagnier-Dauchelle



Thierry Judge



Jose Dolz



Heber Hwang Arcolezi



Sylvain Bouix



Introduction

What will I learn ?

Round table:

- The foreseeable impact of AI on society and medicine: who wins? who loses?

Moderator



Rola Harmouche

Panelists



Laurent Elkrief



Julie Hussin



Maria Cutumisu



Udunna Anazodo



Introduction

How to apply these concepts ?

Four hands-on tutorials:

1. Classification from machine learning to deep learning part 1
2. Classification from machine learning to deep learning part 2
3. Segmentation using deep learning
4. Variational Autoencoder
5. Foundation models

 PyTorch


TensorFlow



Introduction

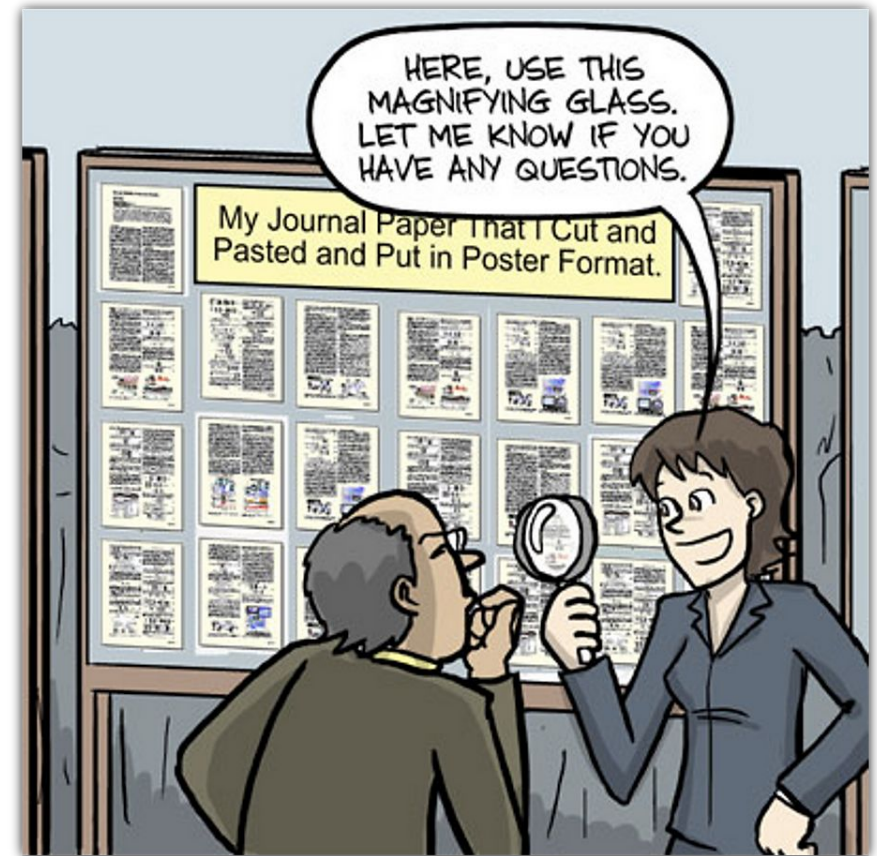
Meeting other participants

Welcome cocktail and Poster session:

Tonight 18h00 - 20h00, Outside this auditorium

Setup your poster early on *ANY* available stand

Bring your drink coupons



Introduction

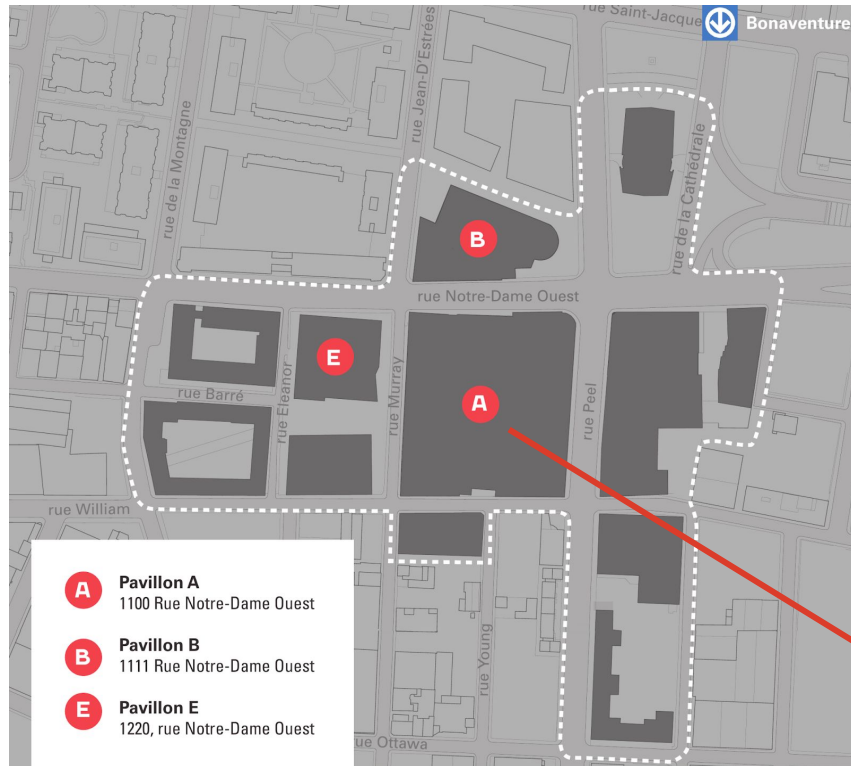
Having fun is also important...

Banquet dinner at Château Ramezay: Thursday 18h30-21h00



Practical information

Campus map



Auditorium (here)

Talks, coffee breaks, posters (outside)

Cafeteria (ground floor)

Lunch

Computer labs (2nd floor)

Hands-on tutorials



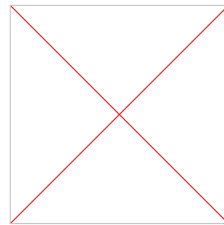
Practical information

Wi-Fi anywhere on campus

Network: **ETS-Invites**

Login: **wifi-even@etsmtl.ca**

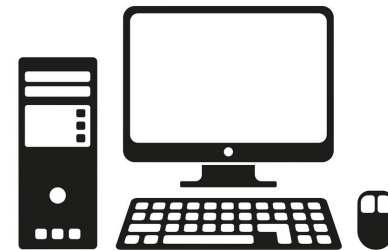
Password: **Eve-Ets!!**



No laptop ? No problem !

Use the same login on any PC in the labs

See booklet for more information



Practical information

Hands-on sessions

Held simultaneously in two different computer labs:

Sessions 1, 2, 3 : Rooms A-2446 and A-2500

Session 4 : Rooms A-2404 and A-2500

Session 5 : Rooms A-2446 and A-2509

You can go to any of the two rooms, we will then assign you to a desk based on available space

Follow signs to find rooms



Practical information

Hands-on material

Documents and videos can be found here:

URL : <https://jodoin.github.io/dlmi2026/>

Password : **dlmi2026**

Before hands-on:

- Create a Compute Canada account (*see web page*)
- Review the material



Practical information

Breakfast and coffee breaks outside the auditorium

- Light breakfast between 8h30 and 9h00
- Coffee breaks at 10h30 and 15h30



Lunch at the ÉTS cafeteria

- Usually between 12h30 and 14h
- **Bring your meal coupons (one per day)**

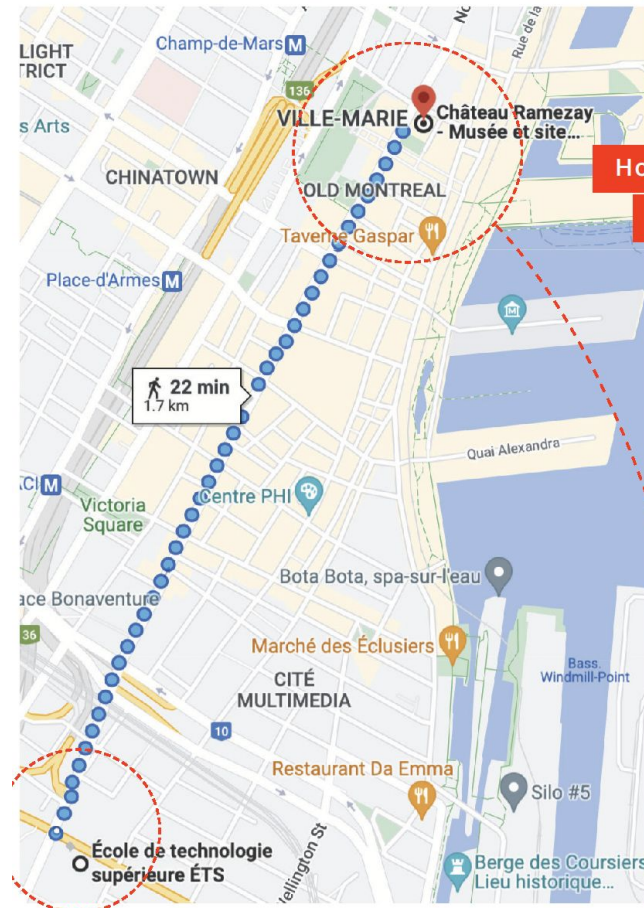


Practical information

Going to the banquet
dinner on Thursday

(*Château Ramezay*)

See map in the booklet:

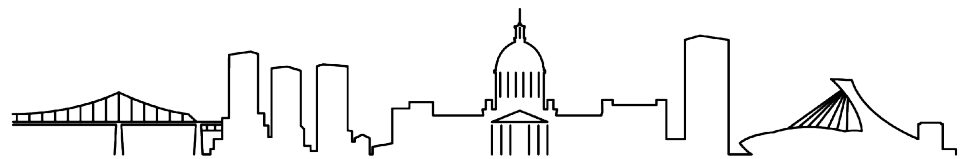


How to get to the
Château Ramezay

A walking group will
leave from the main
entrance of this
building at **18h00**

Feel free to join !



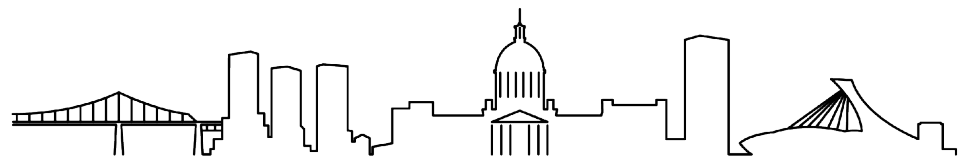


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Thanks for listening
Have a great time !





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Will be back soon... (see schedule)

